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COMMON QUARTERLY EXAMINATION - 2026

Standard XII

Reg.No. | | | | | | |

MATHEMATICS

Time : 3:00 hrs **Part - I** *Marks : 90*

I. Choose the correct answer. **20 x 1 = 20**

1. If $|adj(adjA)| = |A|^9$, then the order of the square matrix A is
   a) 3 b) 4 c) 2 d) 5
2. Rank of the null matrix is
   a) 0 b) 1 c) n d) n + 1
3. If $(AB)^{-1} = \begin{pmatrix} 12 & -17 \\ -19 & 27 \end{pmatrix}$ and $A^{-1} = \begin{pmatrix} 1 & -1 \\ -2 & 3 \end{pmatrix}$, then $B^{-1} =$
   a) $\begin{pmatrix} 2 & -5 \\ -3 & 8 \end{pmatrix}$ b) $\begin{pmatrix} 8 & 5 \\ 3 & 2 \end{pmatrix}$ c) $\begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$ d) $\begin{pmatrix} 8 & -5 \\ -3 & 2 \end{pmatrix}$
4. If $|z - 2 + i| \le 2$, then the greatest value of $|z|$ is
   a) $\sqrt{3} - 2$ b) $\sqrt{3} + 2$ c) $\sqrt{5} + 2$ d) $\sqrt{5} - 2$
5. $\overline{3i} + \frac{i}{2-i}$ is
   a) $\frac{2}{5} - \frac{14i}{5}$ b) $\frac{2}{5} + \frac{14i}{5}$ c) $\frac{-2}{5} + \frac{14i}{5}$ d) $\frac{-2}{5} - \frac{14i}{5}$
6. If α and β are the roots of $x^2 + x + 1 = 0$, then $\alpha^{2020} + \beta^{2020}$ is
   a) –2 b) –1 c) 1 d) 2
7. If –i + 3 is a root of $x^2 - 6x + k = 0$, then the value of k is
   a) 5 b) $\sqrt{5}$ c) $\sqrt{10}$ d) 10
8. A zero of $x^3 + 64$ is
   a) 0 b) 4 c) 4 i d) –4
9. If $x^3 + 12x^2 + 10ax + 1999$ definitely has a positive zero if and only if
   a) $a \ge 0$ b) $a > 0$ c) $a < 0$ d) $a \le 0$
10. The principal value of arg z is
    a) $(-\pi, \pi)$ b) $[-\pi, \pi]$ c) $[-\pi, \pi)$ d) $(-\pi, \pi]$
11. $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right)$ is equal to
    a) $\frac{1}{2}\cos^{-1}\left(\frac{3}{5}\right)$ b) $\frac{1}{2}\sin^{-1}\left(\frac{3}{5}\right)$ c) $\frac{1}{2}\tan^{-1}\left(\frac{3}{5}\right)$ d) $\tan^{-1}\left(\frac{1}{2}\right)$

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12. If $\sin^{-1}x + \cot^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{2}$, then x is

a) $\frac{1}{2}$ b) $\frac{1}{\sqrt{5}}$ c) $\frac{2}{\sqrt{5}}$ d) $\frac{\sqrt{3}}{2}$

13. The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ is

a) 1 b) 3 c) $\sqrt{10}$ d) $\sqrt{11}$

14. The area of the quadrilateral formed with foci of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$ is

a) $4(a^2 + b^2)$ b) $2(a^2 + b^2)$ c) $a^2 + b^2$ d) $\frac{1}{2}(a^2 + b^2)$

15. Area of the greatest rectangle inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

a) 2ab b) ab c) $\sqrt{ab}$ d) a/b

16. If $y^2 = -8x$, then the directrix is

a) $x + 2 = 0$ b) $x - 2 = 0$ c) $y + 2 = 0$ d) $y - 2 = 0$

17. If $\vec{a}.\vec{b} = \vec{b}.\vec{c} = \vec{c}.\vec{a} = 0$, then the value of $[a^{-1}\ b^{-1}\ c^{-1}]$ is

a) $|\vec{a}||\vec{b}|\vec{c}|$ b) $\frac{1}{3}|\vec{a}||\vec{b}|\vec{c}|$ c) 1 d) –1

18. If the direction cosines of a line are $\frac{1}{c}, \frac{1}{c}, \frac{1}{c}$ then

a) $c = \pm 3$ b) $c = \pm\sqrt{3}$ c) $c > 0$ d) $0 < < < 1$

19. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + \hat{j}$, $\vec{c} = \hat{i}$ and $(\vec{a} \times \vec{b}) \times \vec{c} = \lambda\vec{a} + \mu\vec{b}$, then $\lambda + \mu$ is

a) 0 b) 1 c) 6 d) 3

20. Distance from origin to the plane $3x - 6y + 2z + 7 = 0$ is

a) 0 b) 1 c) 2 d) 3

Part - II

II. Answer any Seven questions. (Q.No.30 is compulsory) **7 x 2 = 14**

21. Find the equation of circle with centre (–3,–4) and radius 3 units.

22. Prove that $A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ is orthogonal.

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23. Simplify : $\sum_{n=1}^{10} i^{n+50}$

24. Find the centre and radius of circle $|2z + 2 - 4i| = 2$

25. Find the cubic equation with roots 1, 2 and 3

26. Find period and amplitude of $y = 4 \sin(-2x)$

27. Find the value of $\tan\left[\cos^{-1}\left(\frac{1}{2}\right) - \sin^{-1}\left(-\frac{1}{2}\right)\right]$

28. Find the equation of tangent at $t = 2$ to the parabola $y^2 = 4ax$

29. Find the distance between the parallel planes $x + 2y - 2z + 1 = 0$ and $2x + 4y - 4z + 5 = 0$

30. If A is a non-singular matrix of odd order, prove that $|\text{adj } A|$ is positive.

Part - III

III. Answer any Seven questions. (Q.No.40 is compulsory) **7 x 3 = 21**

31. Find the rank of the matrix $\begin{pmatrix} 1 & 1 & 1 & 3 \\ 2 & -1 & 3 & 4 \\ 5 & -1 & 7 & 11 \end{pmatrix}$

32. Solve by matrix inversion method : $5x - 2y + 16 = 0,\ x + 3y - 7 = 0$

33. Show that the equation $z^2 = \bar{z}$ has four solutions.

34. If $|z| = 2$, then show that $3 \le |z + 3 + 4i| \le 7$

35. Find the value of $\sin\left(\frac{\pi}{3} + \cos^{-1}\left(\frac{-1}{2}\right)\right)$

36. Find the equation of circle whose end points of diameter are (3,4) and (2,–7)

37. Find the equation of tangent and normal to the parabola $x^2 + 6x + 4y + 5 = 0$ at (1,–3)

38. Prove that $\left[\vec{a} + \vec{b}\ \ \vec{b} + \vec{c}\ \ \vec{c} + \vec{a}\right] = 2\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]$

39. Find the distance of a point (2,5,–3) from the plane $\vec{r} \cdot \left(6\hat{i} - 3\hat{j} + 2\hat{k}\right) = 5$

40. If α and β are the roots of the equation $x^2 + 5x + 6 = 0$, then show that $\alpha^2 + \beta^2 = 13$

Part - IV

IV. Answer all the questions. **7 x 5 = 35**

41. a) Find the domain of $f(x) = \sin^{-1}\left(\frac{|x|-2}{3}\right) + \cos^{-1}\left(\frac{1-|x|}{4}\right)$

(OR)

b) By using Gaussian elimination method, balance the chemical reaction equation

$C_5H_8 + O_2 \rightarrow CO_2 + H_2O$

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42. a) Solve: $\tan^{-1}\left(\frac{x-1}{x-2}\right)+\tan^{-1}\left(\frac{x+1}{x+2}\right)=\frac{\pi}{4}$

(OR)

b) Prove by vector method that $\sin(\alpha-\beta)=\sin\alpha\cos\beta-\cos\alpha\sin\beta$

43. a) If $2+i$, $3-\sqrt{2}$ are roots of the equation $x^6-13x^5+62x^4-126x^3+65x^2+127x-140=0$, find all roots.

(OR)

b) If $\cos^{-1}x+\cos^{-1}y+\cos^{-1}z=\pi$ and $0<x, y, z<1$, show that $x^2+y^2+z^2+2xyz=1$

44. a) Solve : $\frac{3}{x}-\frac{4}{y}-\frac{2}{z}-1=0$, $\frac{1}{x}+\frac{2}{y}+\frac{1}{z}-2=0$, $\frac{2}{x}-\frac{5}{y}-\frac{4}{z}+1=0$

(OR)

b) On lighting a rocket cracker, it gets projected in a parabolic path and reaches a maximum height of 4 m when it is 6 km away from the point of projection. Finally it reaches the ground 12 m away from the starting point. Find the angle of projection.

45. a) Show that $x-y+4=0$ is a tangent to the ellipse $x^2+3y^2=12$. And also find the point of contact.

(OR)

b) Find the parametric, non-parametric vector equation and Cartesian equation of the plane passing through the points $(-1,2,0)$ and $(2,2,-1)$ and parallel to the straight line $\frac{x-1}{1}=\frac{2y+1}{2}=\frac{z+1}{-1}$

46. a) If $\vec{a}=\hat{i}-\hat{j}$, $\vec{b}=\hat{i}-\hat{j}-4\hat{k}$, $\vec{c}=3\hat{j}-\hat{k}$ and $\vec{d}=2\hat{i}+5\hat{j}+\hat{k}$, verify that $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d})=[\vec{a}\ \vec{b}\ \vec{d}]\vec{c}-[\vec{a}\ \vec{b}\ \vec{c}]\vec{d}$

(OR)

b) Solve: $6x^4-35x^3+62x^2-35x+6=0$

47. a) If $z=x+iy$ and $\arg\left(\frac{z-i}{z+2}\right)=\frac{\pi}{4}$, then show that $x^2+y^2+3x-3y+2=0$

(OR)

b) A tunnel through a mountain for a four lane highway is to have an elliptical opening. The total width of the highway (not the opening) is to be 16 m and the height at the edge of the road must be sufficient for a truck 4 m high to clear if the highest point of the opening is to be 5 m approximately. How wide must the opening be?

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